



# **CIS 635 Knowledge Discovery & Data Mining**

Features encoding!



# Outline

- Label Encoding
- One-Hot Encoding



# Data/Feature Encoding

| Age<br>(Yr) | Height<br>(M) | Weight<br>(Kg) |
|-------------|---------------|----------------|
| 10          | 1.3           | 45             |
| 30          | 1.7           | 67             |
| 65          | 1.5           | 57             |
| ...         | ...           | ...            |

*Given a Data Table*



# Data/Feature Encoding

| Age<br>(Yr) | Height<br>(M) | Weight<br>(Kg) |
|-------------|---------------|----------------|
| 10          | 1.3           | 45             |
| 30          | 1.7           | 67             |
| 65          | 1.5           | 57             |
| ...         | ...           | ...            |

*Numeric (R) values at  
different scales*

*Given a Data Table*



# Data/Feature Encoding

| Age<br>(Yr) | Height<br>(M) | Weight<br>(Kg) | Gender |
|-------------|---------------|----------------|--------|
| 10          | 1.3           | 45             | F      |
| 30          | 1.7           | 67             | M      |
| 65          | 1.5           | 57             | F      |
| ...         | ...           | ...            | ...    |

*Given a Data Table*

*Categorical  
Feature*

# Data/Feature Encoding

| Age (Yr) | Height (M) | Weight (Kg) | Gender |
|----------|------------|-------------|--------|
| 10       | 1.3        | 45          | F      |
| 30       | 1.7        | 67          | M      |
| 65       | 1.5        | 57          | F      |
| ...      | ...        | ...         | ...    |

Given a Data Table

Categorical  
Feature

$\{F, M\}$

$\{F=0, M=1\}$

# Data/Feature Encoding

*Given a Data Table*

| Age (Yr) | Height (M) | Weight (Kg) | Gender | Color preference |
|----------|------------|-------------|--------|------------------|
| 10       | 1.3        | 45          | F      | Red              |
| 30       | 1.7        | 67          | M      | Blue             |
| 65       | 1.5        | 57          | F      | Green            |
| ...      | ...        | ...         | ...    | ...              |

*Categorical  
Feature*

# Data/Feature Encoding

Given a Data Table

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| ...      | ...        | ...         | ...    | ...              |

Categorical  
Feature

{R, G, B}

{R=0, G=1, B=2}

# Data/Feature Encoding

*Given a Data Table*

| Age (Yr) | Height (M) | Weight (Kg) | Gender | Color preference |
|----------|------------|-------------|--------|------------------|
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| ...      | ...        | ...         | ...    | ...              |

$\{F, M\}$   
 $\{F=0, M=1\}$

$\{R, G, B\}$   
 $\{R=0, G=1, B=2\}$

Label encoder

Categorical  
Feature

# Data/Feature Encoding

*Given a Data Table*

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| ...      | ...        | ...         | ...    | ...              |

$\{R, G, B\}$

$\{R=0, G=1, B=2\}$

Label encoder

Do you see any issue here?

# Data/Feature Encoding

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| ...      | ...        | ...         | ...    | ...              |

- We are saying, Green is closer to Red than Blue,

{R, G, B}

{R=0, G=1, B=2}

Do you see any issue here?

Label encoder

# Data/Feature Encoding

Given a Data Table

| Age (Yr) | Height (M) | Weight (Kg) | Gender | Color preference |
|----------|------------|-------------|--------|------------------|
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| ...      | ...        | ...         | ...    | ...              |

- We are saying, Green is closer to Red than Blue,
- And enforcing Bias in the Vector Space

{R, G, B}

{R=0, G=1, B=2}

Do you see any issue here?

Label encoder

# Data/Feature Encoding

Given a Data Table

| Age (Yr) | Height (M) | Weight (Kg) | Gender | Color preference |
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| ...      | ...        | ...         | ...    | ...              |

- We are saying, Green is closer to Red than Blue,
- And enforcing Bias in the Vector Space
- Which doesn't seem ok.

{R, G, B}

{R=0, G=1, B=2}

Do you see any issue here?

Label encoder



# Data/Feature Encoding

## One Hot Encoding

- *How many colors do we have?*



# Data/Feature Encoding

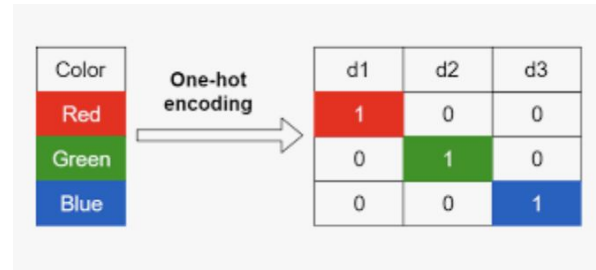
## One Hot Encoding

- *How many colors do we have?*
- 3    {R, G, B}

# Data/Feature Encoding

## One Hot Encoding

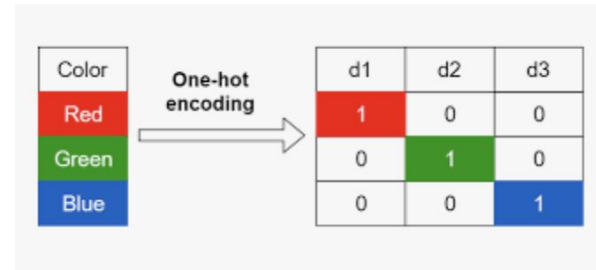
- *How many colors do we have?*
- 3     {R, G, B}
- *So, we use 3 Bits to define each color*



# Data/Feature Encoding

## One Hot Encoding

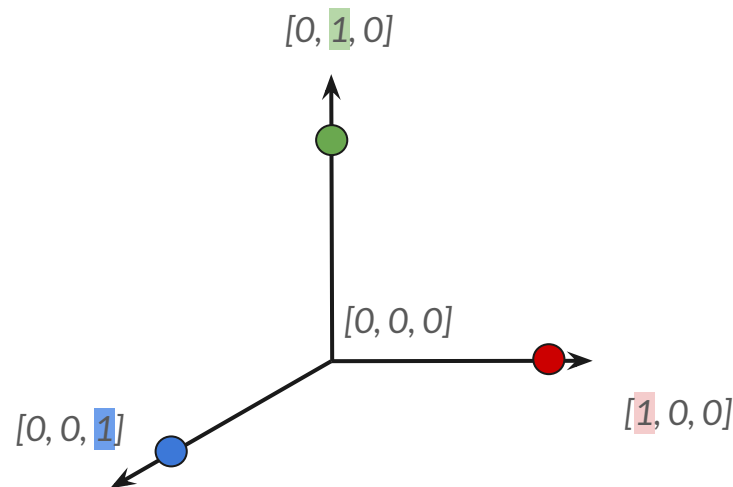
- *How many colors do we have?*
- 3 {R, G, B}
- *So, we use 3 Bits to define each color*
- *How does it solves the Vector Space Distance Problem?*



# Data/Feature Encoding

## One Hot Encoding

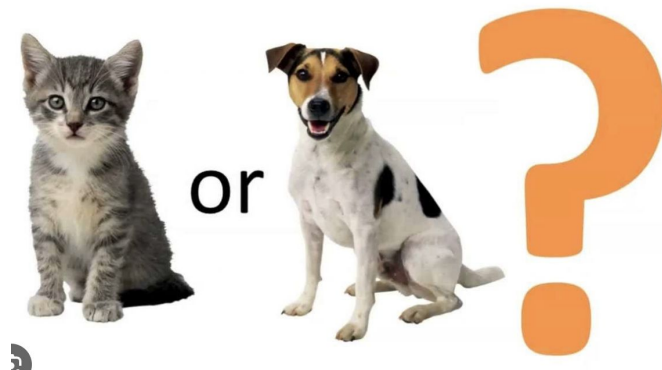
- *How many colors do we have?*
- 3      $\{R, G, B\}$
- *So, we use 3 Bits to define each color*
- *How does it solves the Vector Space Distance Problem?*
- *We have debiased the definition. Isn't it Cool!*



## One hot encoding (cont.)

Classification task:

- Binary example {Cat vs Dog}
- Set size is 2
  - Cat (0, 1)
  - Dog (1, 0)
  - Or vice versa
- Same rule applies every categorical data





# Notebook Presentation!

*Regression task with categorical variables.*



**QA**